

Final Project Report: Analyzing the Insulation Performance of Various Blankets

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Executive Summary

Many blankets on the market advertise themselves as being “thermal blankets” that can retain heat well. However, many of these blankets are made of different materials, which raises the question of which material is most effective for insulating human skin. In this experiment, three different blankets made of polyester, down, and wool were tested to determine their thermal conductivities (k).

Epoxy resin plates of dimensions (6.35 cm x 6.35 cm x 0.381 cm) were heated to 37°C using a patch heater connected to a power stat turned up to 40V. The temperature was measured using thermocouples and the LabView Temperature Logger. After heating, the block was wrapped in a blanket and then placed into a cooler maintained at 5°C. The temperature of the plate was measured for an hour, after which the trial would conclude, and be repeated with a blanket of different material.

With the data collected, a transient analysis was performed. This procedure involved verifying the lumped capacitance model, linearizing temperature vs time curves, and using the slope to determine the overall heat transfer coefficient for each material. This number was used to calculate the final k values.

The lowest k measured was wool, at $0.0152 \frac{W}{mK}$, followed by down at $0.0344 \frac{W}{mK}$, and then polyester at the highest k of $0.390 \frac{W}{mK}$. The lower k values indicate greater performance in insulation and heat retention.

These results were measured against the general published values for the k of each material. The comparison yielded a slightly alarming 67.66% error for wool, a 37.60% error for down, and a reassuring 2.63% error for polyester. Overall, the comparison reflected that the values obtained experimentally were within reason, and the conclusions maintain validity.

Generally, the experimental procedure has room for improvement by revising the simplifying assumptions made, as well as increasing the frequency and range of trials. With that being said, it was still a successful experiment with valuable takeaways. While many companies may claim to have the best insulating blanket, the material used makes a big difference. Polyester is by far poorer than other materials at retaining heat, while down does a better job, and wool is even more effective. This can help educate consumers of such products upon deciding which one to purchase.

Introduction

The project discussed in this report focuses on the analysis of the insulation performance of various blankets on human skin. The objective is to test and evaluate the conductive properties of three blankets made of different materials to determine which is the most effective thermal insulator. The best insulator is determined by which blanket has the lowest thermal conductivity (k). To simulate the thermal properties of human skin, epoxy resin was used. Epoxy resin is commonly used in physical skin models for research purposes as it has similar thermal properties to that of human skin^[1].

The experiment was performed by wrapping the epoxy resin in a blanket, and heating the epoxy resin to a temperature of 37°C, the temperature of human skin, using a patch heater. The heated epoxy resin wrapped in the blanket, was then placed into a cooler maintained at 5°C. The heated epoxy resin would mimic human skin wrapped in a blanket exposed to a cold environment. This would allow us to properly simulate the real-time heat loss of the epoxy resin, showing us how well the blanket allowed the epoxy to retain heat. The temperature of the epoxy was measured every second over a one hour period while it was placed into the cooler.

The data collected during the experiment will then be used to complete an analysis of the thermal conductivity of each blanket material. By comparing the rate of cooling among different samples, the insulation effectiveness of each blanket will be quantified. These experimental results will then be compared to theoretical values to determine the percentage error of the experimental results, recording the accuracy of the experiment itself.

The project aims not only to identify the most effective blanket for thermal insulation but also to test the marketing claims made by manufacturers. Understanding the heat transfer process through different materials can help determine the best popular material for blankets, which has practical applications in military use, outdoor equipment design, and medical care.

Experimental Method

In order to design an experiment for testing the effectiveness of thermal blankets, it must be determined what exactly the blanket will be insulating. This must be something small enough to work with and fit into a mini cooler, and must also be able to house thermocouples to gauge temperatures. Upon review of published literature, it was concluded that epoxy resin would be an appropriate choice. Not only can it be molded in any shape or dimensions needed, it also has very similar thermal properties to human skin, making it a suitable choice for testing blankets that ultimately wrap around human skin. The plate was formed using a silicone mold, having dimensions of 6.35 cm x 6.35 cm x 0.381 cm.

During setup, the plate was heated to a temperature of 37°C using a patch heater on the plate. This temperature was selected to mimic human body temperature. The patch heater was connected to a powerstat that was set to 40V. The mini cooler was plugged in and set to the coldest setting in advance, so that it could reach its steady state of 5°C by the time the plate was heated and ready to go in. The temperatures inside the cooler were gauged using an IR camera.

Due to the nature of the mold, it was not practical to actually insert thermocouples inside of the plate. Therefore, aluminum tape was used to attach a thermocouple to the top of the plate underneath the blanket layer, and then another thermocouple on top of the blanket layer. The blankets were cut into small strips as shown in Figure 1, so that it effectively wraps around the block once. The thickness of each material was (0.635 cm, 0.1016 cm, and 0.23 cm) for the polyester, wool, and down blankets, respectively.



Figure 1: Cut Strips of Polyester, Wool, and Down Blankets (Left to Right)

Once the blanket was wrapped, the block would go into the cooler with the thermocouples connected to LabView temperature logging software. Figure 2 represents the system described.

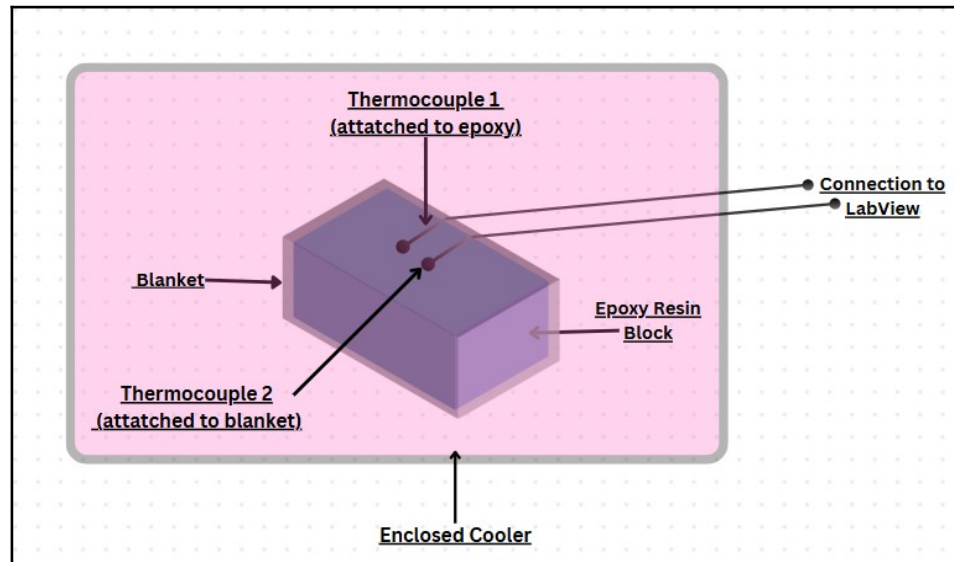


Figure 2: Diagram of Experimental Setup

Each trial was run for an hour, with measurements collected at 1 second increments. Once a trial concluded, the block was heated back to 37°C, and the blanket was replaced with another material for the next trial. Overall, three trials were performed corresponding to three materials tested in the experiment, with temperature vs time data being recorded for each thermocouple in every trial.

Analysis

To evaluate the insulating performance of the different blanket materials, transient analysis based on the lumped capacitance model was used. The goal of this analysis was to determine the thermal conductivity for each of the blankets under the same experimental conditions. The following assumptions were made to help simplify the thermal analysis:

- 1) Unsteady-state heat transfer- The temperature of the epoxy plate changes with time after it is placed in the cooler, making the system transient.
- 2) Negligible radiation- It was assumed that the radiative heat loss was minimal inside the cooler, therefore only conduction and convection were considered

- 3) Uniform heating of the plate- The patch heater was assumed to evenly heat the surface of the plate prior to cooling.
- 4) Uniform thermal properties- Specific heat (C_p), thermal conductivity, and density (ρ) of the epoxy were treated as constants over the temperature range used.
- 5) No internal heat generation- The plate was not generating heat during the cooling phase inside the cooler.
- 6) Negligible contact resistance- The thermal resistance between the epoxy block and the blanket as well as the blanket and cooler was assumed to be minimal.
- 7) Perfect insulation on the sides of the plate- Heat loss through from the sides of the plate were considered negligible. Only heat transfer from the top and bottom faces of the plate were factored into calculations.
- 8) Thermocouples are ideal and provide accurate point measurements
- 9) Ambient cooler air remains constant- The cooler was assumed to maintain $T_\infty = 5^\circ\text{C}$ for the entire duration of the experiment

With these assumptions, the simplified system used for the analysis is highlighted in Figure 3 shown below.

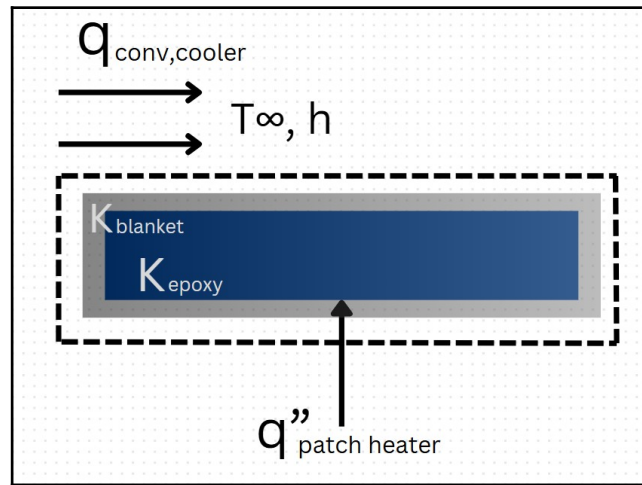


Figure 3: Schematic of the experimental system

The next step in the analysis was to verify the lumped capacitance model. To do this, the Biot number (Bi) must satisfy the following with h_{cooler} being the convection coefficient inside the

cooler, L_c being the characteristic length of the epoxy plate (volume/area), and k_{epoxy} being the thermal conductivity of the epoxy:

$$0.1 > Bi = \frac{h_{cooler} L_c}{k_{epoxy}} \quad (1)$$

With h_{cooler} equal to $2.27 \frac{W}{m^2/K}$ ^[2], L_c equal to 0.001905 m, and k_{epoxy} equal to $0.14 \frac{W}{m \cdot K}$ ^[3], the Bi number was calculated to be 0.03, which is less than 0.1. Therefore, the lumped capacitance model was verified.

Moving forward, during the experiment, data on the temperature (°C) versus time (s) of the epoxy plate was collected. The cooling of the epoxy plate was modeled through the following function with τ being the time constant, $T_\infty = 5^\circ C$, and $T_i = 37^\circ C$.

$$\frac{T(t) - T_\infty}{T_i - T_\infty} = e^{-t/\tau} \quad (2)$$

This equation was then linearized through taking the natural log of both sides:

$$\ln\left(\frac{T(t) - T_\infty}{T_i - T_\infty}\right) = \frac{-t}{\tau} \quad (3)$$

From this, a graph of $\ln\left(\frac{T(t) - T_\infty}{T_i - T_\infty}\right)$ vs t was generated and τ was found to be equal to the negative reciprocal of the slope of the linear region of the graph.

Next, the overall heat transfer coefficient (U) was calculated for each of the blankets using the following equation with m_{plate} being the mass of the plate (18g), C_p being the specific heat of epoxy resin ($1000 \frac{J}{kg \cdot K}$)^[4], and A being the area that heat transfer is occurring ($7.4 \cdot 10^{-3} m^2$):

$$U = \frac{m_{plate} C_p}{A \tau} \quad (4)$$

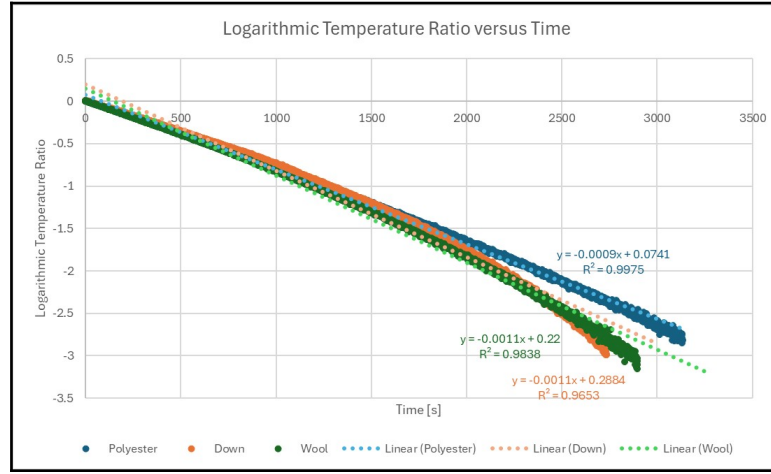
From this, the thermal conductivity for each blanket was solved for by using the total heat transfer equation with L_b being the thickness of each blanket:

$$k_{blanket} = \frac{L_b}{\frac{1}{U} - \frac{1}{h_{cooler}}} \quad (5)$$

Results & Discussion

Figure 4 is

the plot of



$\ln\left(\frac{T(t)-T_{\infty}}{T_i-T_{\infty}}\right)$ vs time for each of the blankets.

Figure 4: Plot of $\ln\left(\frac{T(t)-T_{\infty}}{T_i-T_{\infty}}\right)$ vs time

This plot helps visualize the logarithmic temperature ratio for the epoxy surface wrapped with each of the blankets. The slope for each of the blankets was found to be -0.0009 for polyester, -0.0011 for wool, and -0.0011 for down. For the wool and down blankets, the last 274 seconds and 351 seconds were omitted, respectively. This omission was due to the fact that the

time constant is based on the linear part of the graph and there was a severe non-linear pattern in the data after those times mentioned above. Table 1 shows a summary of the calculated results for each of the blankets.

Blanket	τ (s)	$U \left(\frac{W}{m^2 K} \right)$	L_b (cm)	Calculated $k \left(\frac{W}{mK} \right)$	Published $k \left(\frac{W}{mK} \right)$	% error
Polyester	1111.1	2.19	0.635	0.390	0.380 ^[5]	2.63%
Wool	909.09	2.68	0.1016	0.0152	0.047 ^[6]	67.66%
Down	909.09	2.68	0.23	0.0344	0.025 ^[7]	37.60%

Table 1: Summary of calculated results

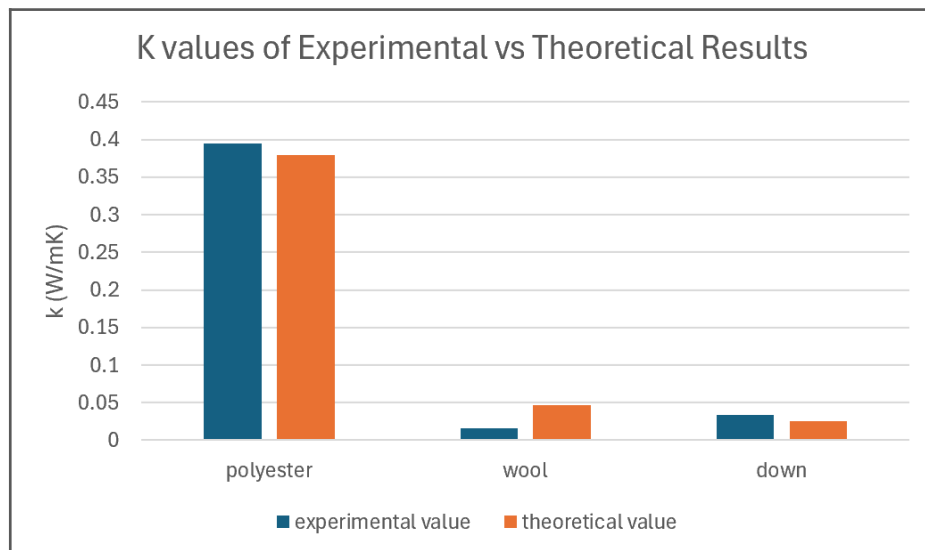


Figure 5: Comparing the theoretical and experimental k values

Conclusions & Recommendations

The experiment sought to identify which of three materials (polyester, down, wool) forms the best blanket for thermal insulation. This was done by experimentally determining the thermal conductivity of each material through a series of trials where each blanket's heat retention capabilities were put to the test under the same conditions of being wrapped around an epoxy plate while placed in a cooler. Subsequently, transient analysis was used to determine that wool

was the most effective insulator, followed by down, and with polyester having significantly poorer performance. The detailed results are highlighted above in Table 1.

The accomplishments in this experiment are as follows:

- Epoxy resin plates were successfully formed to mimic human skin
- Plates were appropriately heated to 37°C to match neutral human body temperature
- A steady state temperature value was correctly maintained in the cooler
- Temperature trends were effectively measured with thermocouples
- The lumped capacitance model was verified
- Transient analysis produced realistic thermal conductivity values
- Comparison to theoretical published values matched experimental results

Overall, the results achieved the goal of understanding the significance of blanket material composition and gave insight into which materials perform best. In the future, this experiment can be extended by testing even more materials beyond the three trialed in this experiment in order to arrive at a more comprehensive, absolute conclusion of which materials perform best. Additionally, given more time, performing more than one trial for each experiment could produce more reliable results. Another worthwhile extension to the experiment would be to upgrade the model of the human body from a simple epoxy plate in ways that could mimic the various layers of the skin, blood flow, hair, and even the constant heat generated by it.

References

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Appendix

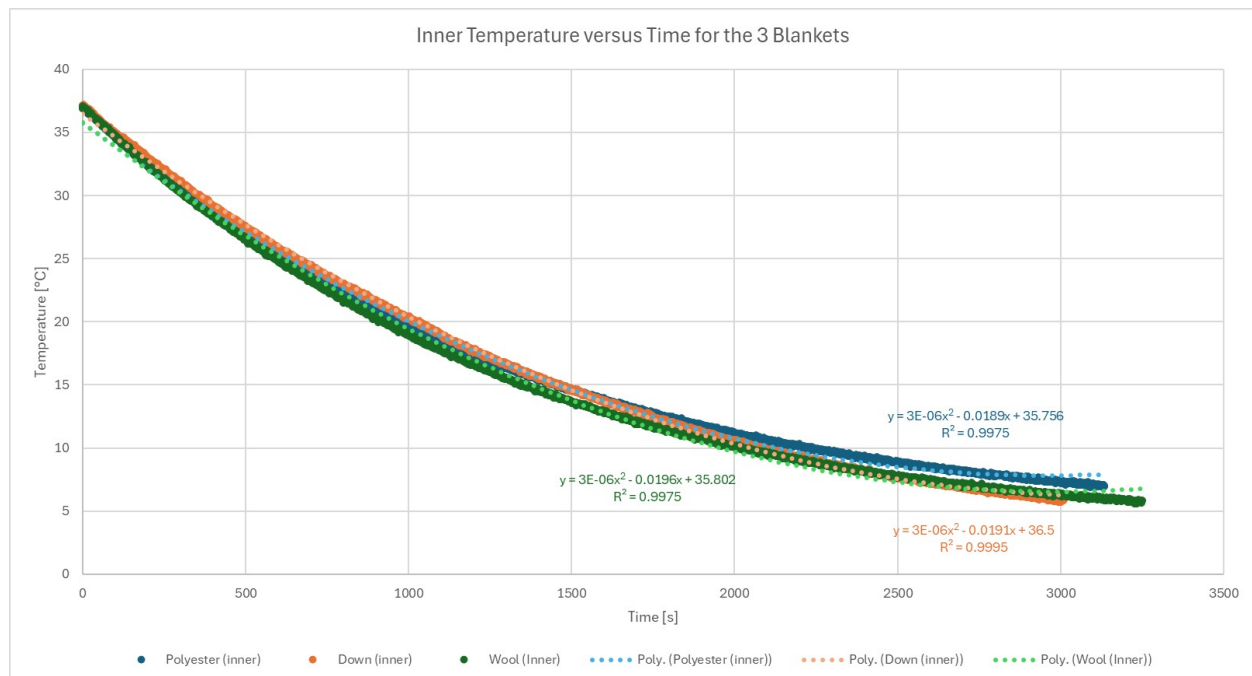


Figure A1: Temperature vs time graph

$$Bi = \frac{L_c \times h_{\text{refrigerator}}}{k_{\text{polyester}}} = \frac{6.35 \times 6.35 \times 0.381 \times 10^{-6}}{6.35 \times 6.35 \times 10^{-4}} \times 2.27 \times \frac{1}{0.14} = 0.03 < 0.1 \quad \checkmark$$

$$\frac{T(t) - T_{\infty}}{T_i - T_{\infty}} = e^{-t/\tau} \Rightarrow \tau = \frac{1}{\text{slope}} \text{ of } \ln \left(\frac{T(t) - T_{\infty}}{T_i - T_{\infty}} \right) \text{ vs time plot}$$

Polyester:

$$\tau_{\text{polyester}} = \frac{1}{0.0009} = 1111.11 \text{ s}$$

$$U_{\text{polyester}} = \frac{m_{\text{plate}} \times C_p}{A \times \tau} = \frac{0.018 \times 1000}{7.4 \times 10^{-3} \times 1111.11} = 2.189 \text{ W/m}^2\text{K}$$

$$k_{\text{polyester}} = \frac{L_{\text{polyester}}}{\frac{1}{U_{\text{polyester}}} - \frac{1}{h_{\text{refrigerator}}}} = \frac{0.00635}{\frac{1}{2.189} - \frac{1}{2.27}} = 0.390 \text{ W/m.K}$$

Wool:

$$\tau_{\text{wool}} = \frac{1}{0.0011} = 909.09 \text{ s}$$

$$U_{\text{wool}} = \frac{m_{\text{plate}} \times C_p}{A \times \tau} = \frac{0.018 \times 1000}{7.4 \times 10^{-3} \times 909.09} = 2.676 \text{ W/m}^2\text{K}$$

$$k_{\text{wool}} = \frac{L_{\text{wool}}}{\frac{1}{U_{\text{wool}}} - \frac{1}{h_{\text{refrigerator}}}} = \frac{0.001016}{\frac{1}{2.676} - \frac{1}{2.27}} = 0.0152 \text{ W/m.K}$$

Down:

$$\tau_{\text{down}} = \frac{1}{0.0011} = 909.09 \text{ s}$$

$$U_{\text{down}} = \frac{m_{\text{plate}} \times C_p}{A \times \tau} = \frac{0.018 \times 1000}{7.4 \times 10^{-3} \times 909.09} = 2.676 \text{ W/m}^2\text{K}$$

$$k_{\text{down}} = \frac{L_{\text{down}}}{\frac{1}{U_{\text{down}}} - \frac{1}{h_{\text{refrigerator}}}} = \frac{0.0023}{\frac{1}{2.676} - \frac{1}{2.27}} = 0.0344 \text{ W/m.K}$$

Figure A2: Calculations of time constant (τ), heat transfer coefficient (U), and thermal conductivity (k) of each blanket